

1. (+20) **Length of a graph** of $f(x)$ on $[a, b]$ is $L = \int_a^b \sqrt{1 + (f'(x))^2} dx$. Given $f(x) = \ln(x) - \frac{1}{8}x^2$ on $[1, 6]$. (a) Find the length of f on the interval analytically. You will need this to calculate the error in your codes. (b) Write a C++ code parallelized using the **C++ threads** library that numerically evaluates the length of this function on the interval using Riemann sum. Your code should accept the number of points used to partition the interval and the number of threads to spawn. (c) Plot the strong scaling efficiency on 1,2,4,8,16 threads of your code for $n = 1.e8$ partition points, i.e. time versus thread count. (d) Plot the *log* of the numerical error for 10, 100, ...1.e6 partition points (increasing by factors of 10x each time). Submit your code and plots.

1-a)

$$f'(x) = \frac{1}{x} - \frac{x}{4}$$

$$L = \int_1^6 \sqrt{1 + (f'(x))^2} dx$$

$$(f'(x))^2 = \frac{1}{x^2} - \frac{1}{2} + \frac{x^2}{16}$$

$$1 + (f'(x))^2 = \frac{1}{x^2} + \frac{1}{2} + \frac{x^2}{16} = \left(\frac{1}{x} + \frac{x}{4}\right)^2$$

$$\sqrt{1 + (f'(x))^2} = \frac{1}{x} + \frac{x}{4}$$

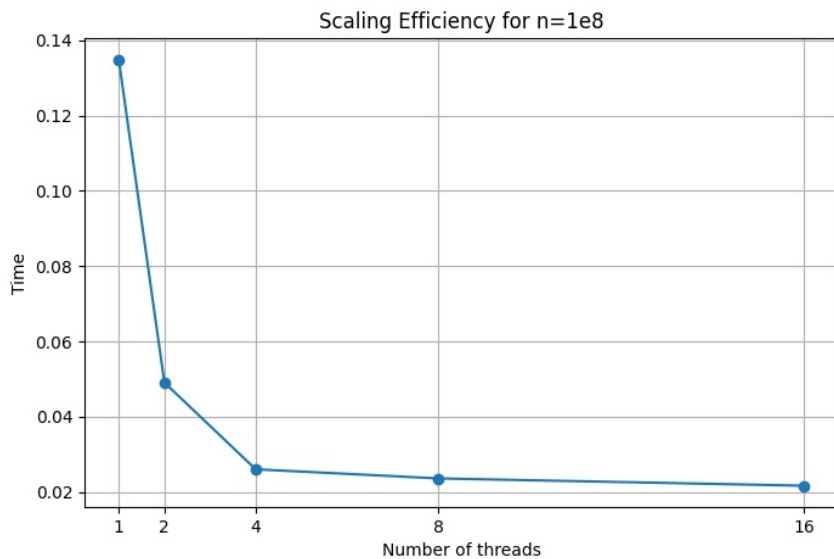
$$L = \int_1^6 \left(\frac{1}{x} + \frac{x}{4}\right) dx = \left[\ln x + \frac{x^2}{8}\right]_1^6 = \ln 6 + \frac{36}{8} - \left(\ln 1 + \frac{1}{8}\right) = \ln 6 + \frac{35}{8}$$

$$\frac{35}{8} = 4.375$$

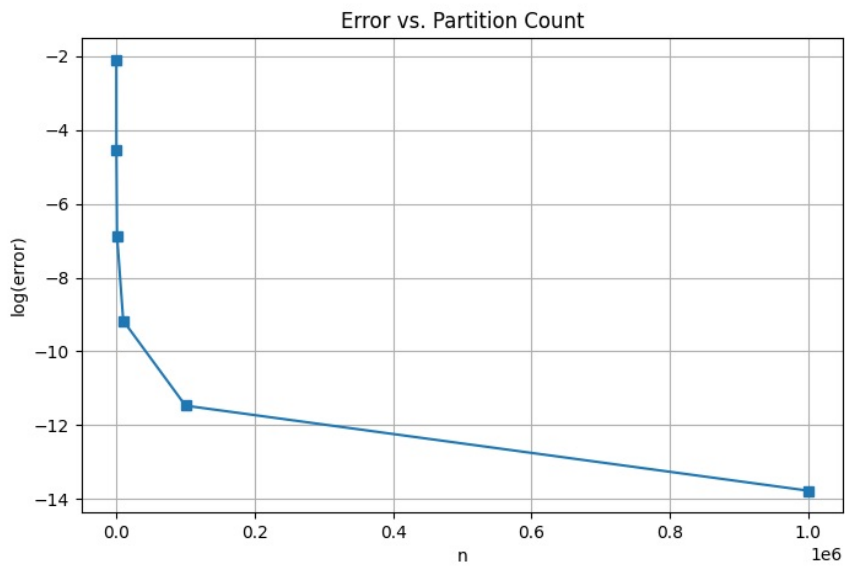
$$\ln 6 \approx 1.791759469228055$$

$$L = \frac{35}{8} + \ln 6 \approx 6.166759469228055$$

1-c)

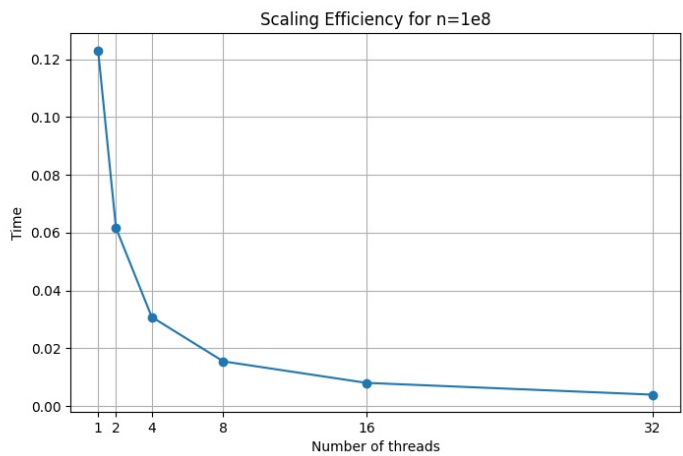


1-d)

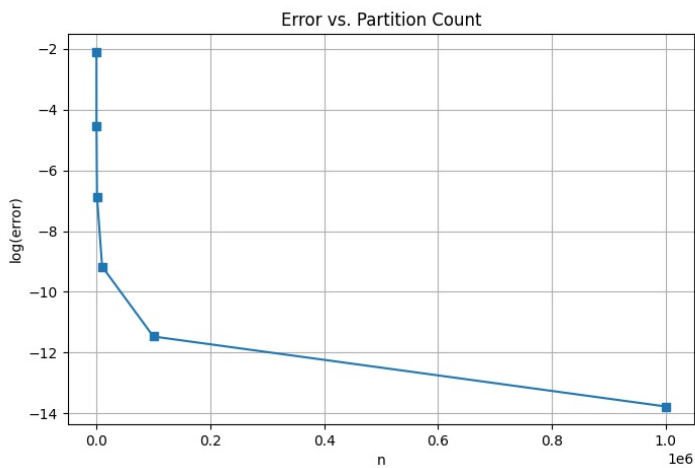


2. (+15) **Length of a graph** of $f(x)$ on $[a, b]$ is $L = \int_a^b \sqrt{1 + (f'(x))^2} dx$. Given $f(x) = \ln(x) - \frac{1}{8}x^2$ on $[1, 6]$. (a) Write a C++ code parallelized using **MPI** that numerically evaluates the length of this function on the interval using Riemann sum. Your code should accept the number of points used to partition the interval. (b) Plot the strong scaling efficiency of your code on 1, 2, 4, 8, 16, 32 processes for $n = 1.e8$ partition points using Hyak. (c) Plot the *log* of the numerical error for 10, 100, ...1.e6 partition points (increasing by factors of 10x each time). Submit your code and plots.

2-b)



2-c)



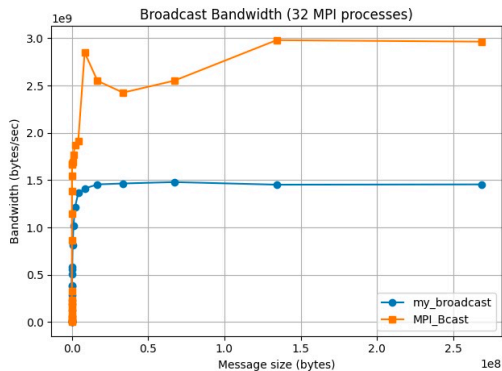
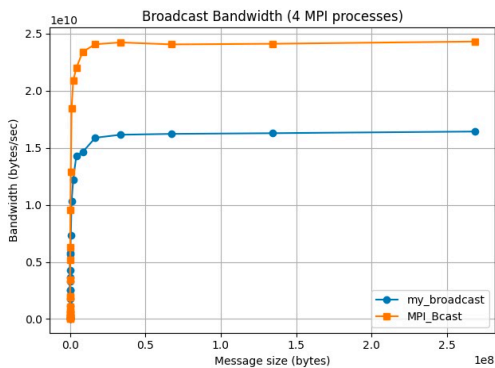
4. (+20) **Communication Bandwidth.** (a) Use the following template declaration to write a **MPI C++** function using point-to-point communication that implements broadcast. Submit the function in file `my_broadcast.hpp`. (b) Measure and plot the performance of your function and the `int MPI_Bcast(void *buffer, int count, MPI_Datatype datatype, int root, MPI_Comm comm)` function on messages of size $8B, 16B, \dots, 256MB$ bytes (doubling the message size each instance). Do this exercise using Hyak on 4 and 32 MPI processes, and make a plot of your measurements with bandwidth (bytes per second) on the y-axis, and the message size in bytes on the x-axis. Submit your plot.

```

• template <typename T>
  void my_broadcast(T* data, int count, int root, MPI_Comm comm);

```

4-b)



In log-scale

